ANBA - Cleaning & Encoding

Page No.:
Date:
M T W T F S S

* 5 Numeric & 7 Categorical columns
* Item_outlet_Sales is DV / Target
* As DV is numeric it is Regression
* Cleaned Item_Fatcontent had messy Values
* Item_Visibility min is 0 Which is not possible so converting 0 to nan

[9..10] missing
* Item_Weight = 1463 , 17% ; median Imputation
Item_Visibility = 526 , 6% ; median Imputation
Outlet_size = 2410 , 28% ; missing are assumed to be Small

[12,13] outliers
* Using 2 methods - IQR & 3 sigma
* Out of 4 num col Visibility (IV) & Sales (DV) had outliers but remain due to real store behaviour

[14] univariate
* num col use histogram & cat col use count plots
* Sales & Visibility are right skewed
* ~~Sales~~ weight & Visibility have a spike at their median due to our median imputation
* Most rows come from Type 1 supermarkets

[15.. Bivariate
* Look at each IV against the DV to find what drives sales using a heatmap
* MRP - Sales = +0.57 ; high correlation
* VIF: all num inputs have VIF ≈ 1 indicating negligible multicollinearity among them
* Scatterplot between numerical and sales
  - MRP shows a Positive relationship (higher priced products tend to generate higher sales Revenue)
  - Visibility has weak negative relationship
  - Weight shows almost no relationship

Quick Work

Page No.:

Date:

M T W T F S S

BoxPlot

* Compare sales accross categorical variables.
  Store type has the strongest relationship
* method: groupby() + median(), compares typical sales
  across categories
  - Fat content: Low Vs Regular has small difference
  - Product type: Relative small difference
  - Outlet type: Strongest sales difference

encoding
* Drop ID's before baseline encoding
* Model compatibility: linear Regression needs numerical inputs
  so all categorical features must be converted
* Why One hot encoding for all cat col?
  to avoid imposing arbitary numerical order or equal
  spacing between categories even Outlet size is one
  hot encoded despite being ordinal
  Outlet_Establishment_Year was also one hot encoded
  because it's treated as catagorical store identifier not
  a continuos numerical variable because it had only
  9 distinct Year accross 10 stores
* 6 cat cols → 31 one hot encoded cols